



# RAP: A Conceptual Business Intelligence Framework

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## ABSTRACT

Over last few years Data Warehouse (DW) has emerged as the core of the Business Intelligence (BI) infrastructure for many forward-looking organizations. It is often complemented by Business Performance Management (BPM) and workflow management tools for operationalizing of strategic and tactical decisions and tracking progress of activities across the organization. However, till now there is not much support available to the decision-makers to leverage “organizational experience” i.e., the past events, corresponding responses made as well as their respective impacts, to guide him/her in the decision-making process. In this paper a conceptual BI framework, the Reference-Activity-Projection (RAP) is proposed. RAP is designed to support the decision-making process through a systematic access mechanism to organizational experience. RAP uses along with standard BI tools and techniques, the emerging technologies such as complex event processing (CEP), unstructured/semi-structured data mining, social network analysis etc.

## Categories and Subject Descriptors

H.3.3 [Information Storage and Retrieval]: Information Search and Retrieval – *Information filtering, Search process, Selection process*

## General Terms

Management, Measurement

## Keywords

Business intelligence, data warehousing, data mining, organizational experience, event processing, social network analysis

## 1. INTRODUCTION

A well-built and well-maintained DW serves as a rich and integrated source of high quality data. Analysis of this data results in developing useful information/knowledge which is of immense importance in decision-making scenario. However, the current

practice usually does not allow systematic maintenance and re-use of these higher level information items. Their availability for future study and re-use can be of immense value. For example, consider a decision-maker trying to find a good solution of a problem. She may very naturally ask the following questions:

- ☐ Are there similar situations those the organization faced in past?
- ☐ What was the response then?
- ☐ Why was that particular response chosen?
- ☐ What was the desired outcome/impact?
- ☐ What was the actual outcome/impact?
- ☐ Why did they differ?
- ☐ What might happen if a similar decision is taken now?
  - How does the business condition differ?
  - How does the market condition differ?
  - What the relative positions are in terms of scopes, deployable resources etc.?

Evidently, answers to such questions require working with information elements at higher level than is possible only with data. We need to maintain and work also with the *instances of knowledge/information developed in past along with their interrelationships*.

Here we call the knowledge elements *Intelligence-Events* or simply *events* and define “organizational experience” as follows:

*Organizational experience is represented by the archived information on past events and their interrelationship. The information about a past event includes the following:*

- The semantic (ontology based) description of the event.
- The actions (decisions) taken in response to the event.
- The estimated impact of implementing the actions.
- The work plan for implementing the decisions.
- The actual impact of the decisions as captured later.
- Various constraints influencing the response and its outcomes.
- The context of the event with respect to other events and data.
- The relationship of the event with other events.

Thus, conceptually, the collection of events form a *network* or *virtual community* of events, where they are linked by various contextual relationships.

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Compute 2008, Jan 18-20, Bangalore, Karnataka, India

© 2008 ACM ISBN 978-1-59593-950-0 /08/01...\$5.00

In this paper a new conceptual BI framework called Reference-Activity-Projection (RAP) framework is proposed. It emphasizes systematic utilization of organizational experience while retaining the standard functionalities of a BI environment. RAP brings together into the context of BI environment several techniques developed in other fields, e.g., complex event processing (CEP) [7] social network analysis (SNA) [10] etc. RAP also uses advanced data mining techniques extensively, especially with semi-structured data such as graphs. In the rest of the paper I elaborate the RAP framework.

Note that, in this paper I do not discuss explicitly the existing technologies of DW and other BI components. Nevertheless, the availability and support of standard technologies, especially extensive *metadata coverage* is assumed implicitly. Further, in this paper focuses on the generic technological aspect of the framework, independent of any particular application domain (e.g., banking, telecom etc.). Naturally, the framework can be adapted for any domain-specific implementation.

## 2. Reference-Activity-Projection (RAP)

The conceptual architecture of the RAP framework is depicted in Figure 1. As the name suggests, the framework is consisted of three modules, **Reference**, **Activity** and **Projection**. However, on the outset it should be stressed that this division is not intended to be translated directly into the logical or physical level design of the DW. For example, the historical data and the current data may occupy same set of tables in the DW data model. Neither does it reflect the division of BI environment into parts such as past, present and future. Here the master data, policy, business rules etc. though form a part of the reference, control the performance of the whole environment. These three modules are described in details in the following subsections.

### 2.1 The Reference

The Reference is the archive of data and information elements. They can be roughly divided into two parts, (1) *data repository* and (2) *event repository*. However, the elements stored in one part can refer to those in the other part. The elements such as historical data, master data, policy, business rules, budget etc. form the data repository. It is evident that from technological viewpoint standard RDBMS possibly with support for simple objects can be used for this. Also standard techniques of data modeling, indexing and querying as developed in the field of DW will work very well here. This part forms the context within which the data analytics take place. They are also accessible to the decision maker through the event objects which used them.

Additionally, to support the RAP architecture effectively, the database and the metadata repository should be designed to rigorously capture and/or preserve the changes over time in various changeable entities like policy, business rules etc. This will allow the decision-maker to put the past events and decisions in proper perspective and study the changes in the business environment.

The other part of Reference, the event repository, deals with the event data and captures the organizational experience. The event data elements are fairly complex objects and managed using the complex event processing (CEP) framework [7] developed by Luckham and co-workers. Though CEP is designed for

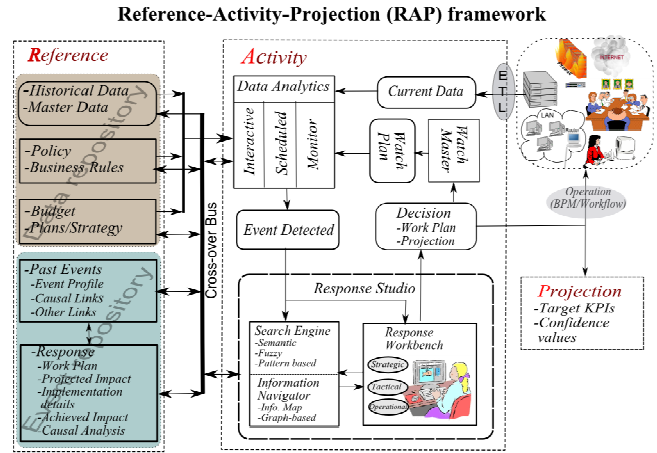


Figure 1. RAP framework

operational systems, the concepts can be adapted easily to the BI usage. The following are the basic concepts in CEP as defined in [7]:

In CEP an “event” is an object that can be subjected to computer processing. It signifies, or is a record of an activity that has happened. An event may be related to other events. An event has three aspects.

- Form: The form of an event is an object. It may have particular attributes or data components.
- Significance: An event *signifies* an activity. We call the activity the significance of the event. An event’s form usually contains the data describing the activity it signifies.
- Relativity: An activity is related to other activities by *time*, *causality* and *aggregation*. Events have same relationships to one another as the activities they signify. The relationship between an event and other events are together called its *relativity*. An event’s form usually encodes its relativities - that is, contains methods that can be invoked to reconstruct the relationships with other events.

Now let us try to adapt the above concepts in BI context. Evidently the CEP, in its original form, is targeted to the (possibly distributed) operational environment. The decision support environment lacks the real-time characteristic of the above. Nevertheless, the following conceptual parallels can be drawn:

The *analytical observations* as well as various *responses* considered and *decisions* taken by the decision-maker can be modeled as *events* in the BI environment, which will henceforth be called *BI event* when necessary for distinction. While the observations can be treated as *external* events, others can be considered as *internal* or *inferred* events. The *aspects* of these events can be reformulated as follows:

- Form: The form of an (BI) event is an *object* with attributes and data components including the *significance* and relativity information.
- Significance: The *significance* of the external events is their *data context*, the constraints used in the analytics and other events such as earlier decisions which have been operationalized and whose results are captured through the system and contributed to the observation. For the internal events the significance is constituted of the set of external

events, past events, policy and other business constraints influencing it and the relevant data context.

- Relativity: BI events are related in many ways including three relationships *time*, *causality* and *aggregation*. They carry time-stamp; they are related causally to others. Most important, BI events naturally form *aggregation hierarchies* where a higher level *complex event* signifying a *broader context* encompasses a set of smaller events.

From above it can be readily seen that most of the concepts in CEP, e.g., event pattern, event network, event hierarchy etc. can be utilized in BI environment also. Event pattern modeling language such as EPLs (Event Pattern Languages) and pattern matching operations [7] also may prove useful for handling of BI events.

## 2.2 The Activity

The Activity encompasses the computational/processing activities in the framework. It can be divided into three sub modules, *Data Analytics*, *Response Studio* and *Watch Master* forming a processing loop. They are discussed in detail in the following.

### 2.2.1 Data Analytics

In RAP the data analytics is consisted of the analytic tools (OLAP, data mining etc.) for performing the analytic tasks using the recently extracted data as well as the reference data. The analytic tasks are coordinated through the specification provided in the *watch-plan*. In other words, the watch-plan invokes the analytic tasks as per requirement specified through it and encodes the results as *observation* events. Again, the items in watch-plan are associated with work-plans. Thus, it creates a *tighter feedback loop* by associating observations with related business actions operationalized earlier. Various significance and relativity information are attached to the observation events at this stage from the data perspective as well as event perspective. The former comes from the analysis that resulted in the event while the event perspective comes through the watch plan(s) which designed the analytics and is itself linked to the events. The event creation by the analytics is depicted in the Figure 2. The output events may be simple ones as well as complex events generated as results of aggregation operations specified by the watch plans. Also it will be helpful at the later stage if *semantic profiles* of the events are built at this stage and attached to the events.

### 2.2.2 Response Studio

This is the nerve center of the whole infrastructure and the place where the decision-maker works with higher level information elements encoded and stored as events as well as data when required. A typical working sequence in response studio may be like as follows:

- Studying the events received from the data analytics involves:
- Trying to formulate a decision to *take advantage of the opportunity* the event presents or to *counter the threat* indicated by the event. Which in turn involves:
    - Using the organizational experience stored in the event repository and other reference data sources.
    - Likely to consider several alternative responses and decide upon the most suitable of them.
    - Preparing a work-plan for implementing the decision

### Creation of an external (observation) event

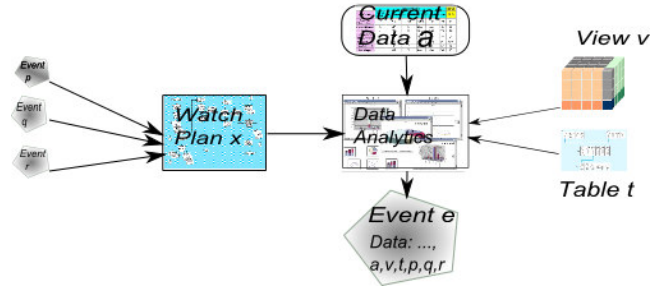


Figure 2. While creating an event object, its significance and relativity information is attached with it..

- Estimating the impact of the decision on the business, express them as target values of various KPIs.
- Creating a decision event and related objects.
- Submitting the decision to a BPM or workflow management tool for operationalization.
- Submitting the decision to the watch master for generating watch plan for tracking the implementation of the decision.
- Submitting the decision to the Projection to be integrated with other projections.

In the following we discuss the issues encountered in accomplishing the above scenario.

As depicted in Figure 1, the decision maker is aided by two tools for working with a mass of events stored in the Reference. The first one, the event *Search Engine*, is used for similarity-based search of the event database. The events being complex and multi-faceted objects, one needs advanced search capability based on data mining techniques for complex objects. This is a very active area of research. Among the available approaches ontology-based semantic search technique [3] and fuzzy or fuzzy-semantic search techniques [1] show much promise. They are likely to mature to a sufficient degree in near future. The RAP framework can be designed to take advantage of these techniques extensively. For this in the previous section I proposed semantic profiling the events. For semantic profiling and search of the objects, it is convenient to use the semantics based on ontology. However, to be really effective the ontology may require being organization-specific [8] and separate effort of building it may be required for different implementations.

The other tool proposed in the framework, the *Information Navigator* is for mining the events based on relationships. Though the CEP framework [7] provides some methods of processing the events based on the relativity, here we need a more powerful set of methods for in-depth analysis of the multidimensional event relationships. Such a collection of events, along with their relationships can naturally be represented as a graph or network. The study of Social Networks, comprised of actors and their interrelationship, has been used for solving a myriad of problems in social science, epidemiology and many other fields [10]. There is a very rich body of techniques developed for analyzing the

networks. In our problem we can consider the collection of events to form a *virtual society/community* and represented in form of graph(s). Along with various data mining techniques for graph data [6][5], the methods of social network analysis can be applied to them for analysis, navigation and visualization. In the information navigator we can envisage the capability of producing a multidimensional map in form of a graph, where the relationship types (such as causality, temporality etc.) will form the dimensions. We can also conceive various aggregation-based OLAP-like operations such as roll-up, drill-down, slicing etc. as well as path-based graph traversal operations.

The response studio generates decisions, corresponding implementation plans and projected impacts. The operationalization of a decision is modeled as an internal event with the other details forming its data components. Further other relevant events studied/used in relation to the decision making process, constitute the relativity of the event. The techniques used in BPM for designing/defining a process [11] can be borrowed for expressing the work plan. This expression can easily be made compatible with the BPM or workflow tools for implementing the process. After the business process is designed an impact analysis can be carried out resulting in a set of projected KPI values and a set of projected events signifying *critical points* during the implementation of the process.

### 2.2.3 Watch Master

The watch master works with a decision object and generates the watch-plan to be implemented by the data analytics. Given a decision event, the corresponding work plan and the projected impact form the basis of the watch plan. This module needs to implement an interface for analyzing various aspect of the decision object and translate then into the components of the watch plan. The watch plan components can include periodic computation of relevant KPIs, comparison with the projected values and generating alarm if they mismatch substantially. On the other end of complexity it can generate plans for detecting complex inferred events employing a series of analytic services. Evidently this is a complex task to achieve. Fortunately this problem has received attention of both academia [9] in different contexts for a long time. The recent trend is to deploy multi-agent expert systems [4] to specify and implement complex tasks. In a recent paper Ali et al. [2] proposed a novel agent based framework for constructing rules and deploying them vis-à-vis business processes. The concepts developed in these works can be incorporated to implement the watch master module.

## 2.3 The Projection

The Projection retains the overall view of the future business conditions. It is comprised of estimated values of various KPIs along with their interrelationship and hierarchical structures. There are also confidence values associated with the KPIs those signify the possibility of achieving the target values of the respective KPIs. The Projection also maintains the projected events whose occurrence signifies milestones in various activities. As a whole, the Projection provides a holistic view of the business in some time in future. It can be subjected to strategic review and fine-tuning of the work plans in a concerted way.

The severest challenge in implementing the above arises from the fact that the KPIs are by nature business domain specific as well as to a large extent organization-specific. Their definition, both

computationally and semantically, may change and evolve with time even within the same organization. In that sense the KPIs are dynamic entities. The same observation applies to a even larger degree to the interrelation among the KPIs. To the best of the knowledge of the author no public literature, addressing this issue satisfactorily and/or generically, is available. It is likely to be implementation specific depending heavily on the input provided by the domain experts and the business experts.

## 3. Conclusion

In this paper I describe a vision for a future generation BI system. I proposed the Reference-Activity-Projection framework with specific aim of aiding the decision-maker through an efficient access to the organizational experience of decision-making. Here I explored various technologies and techniques, even traditionally used in other fields of application, both in existence as well as in nascent stage, and those can be useful in realizing the proposed RAP framework. I have also identified various research issues involved in its realization. Currently research is being carried on to address these issues.

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